Bridgeless PFC Converter

Introduction

In the realm of power electronics, achieving a high power factor is crucial for efficient energy utilization and compliance with regulatory standards. Traditional boost Power Factor Correction (PFC) converters, while effective in maintaining a constant output voltage and drawing unity power factor current, often suffer from inherent limitations. These include a relatively high component count and significant conduction losses due to multiple semiconductor devices conducting simultaneously. This lecture delves into advanced PFC topologies, specifically bridgeless PFC converters, which are designed to overcome these challenges. By reducing the number of conducting devices and optimizing the circuit architecture, bridgeless PFC converters offer improved efficiency and reduced complexity, making them vital for modern charging infrastructure and other power applications.

Core Concepts

Term	Definition	Significance
`Boost PFC Converter`	A power converter that uses a boost DC-DC stage preceded by a full-bridge diode rectifier to achieve power factor correction and output voltage regulation.	Serves as the foundational PFC topology, but its limitations (component count, conduction losses) motivate the development of bridgeless variants.
`Full-Bridge Diode Rectifier`	A circuit comprising four diodes arranged to convert an AC input voltage into a pulsating DC voltage.	Essential component in traditional boost PFC, but contributes to conduction losses and component count.
`Conduction Losses`	Power dissipated as heat in semiconductor devices (diodes, switches) when current flows through them.	A critical metric for converter efficiency; reducing these losses is a primary goal of bridgeless topologies.
`Switching Frequency (fsw)`	The rate at which semiconductor switches (e.g.,	Higher switching frequencies allow for smaller inductors

Term	Definition	Significance
	MOSFETs) are turned on and off in a power converter.	and capacitors but can increase switching losses.
`Line Frequency (fs)`	The frequency of the AC input voltage, typically 50 Hz or 60 Hz.	Diodes switching at line frequency generally have lower losses compared to those switching at high frequency.
`Duty Ratio (d(t))`	The fraction of a switching period during which a switch is turned on. It varies over the line cycle to achieve PFC.	Crucial control parameter for regulating output voltage and shaping input current in PFC converters.
`Volt-Second Balance`	A principle stating that the average voltage across an inductor over one switching period must be zero in steady state.	Used to derive the relationship between input/output voltages and the duty ratio in boost-derived converters.
`Fast Recovery Diode`	A diode designed to switch from conduction to blocking state very quickly, minimizing reverse recovery time and losses.	Necessary for diodes operating at high switching frequencies, but can be costly and contribute to losses.
`Totem Pole PFC`	A specific bridgeless PFC topology that arranges switches and diodes to ensure only active switches operate at high frequency, while diodes operate at line frequency.	Represents a significant advancement in bridgeless PFC, offering high efficiency by minimizing high-frequency diode losses.

Detailed Analysis

1. Limitations of the Conventional Boost PFC Converter

The conventional boost PFC converter, a widely adopted solution for power factor correction, consists of a full-bridge diode rectifier followed by a boost DC-DC converter. While effective, its architecture presents several drawbacks:

- **High Component Count**: It typically requires four diodes for the rectifier, two active switches (S1, S2), an inductor (L), and a capacitor (C). This increases the overall bill of materials and circuit complexity.
- **Significant Conduction Losses**: During operation, at any given instant, three semiconductor devices are typically in conduction. For example, in the positive half-cycle, diode D1, switch S1 (or S2), and diode D4 might be conducting. This simultaneous conduction of multiple devices leads to higher power dissipation and reduced efficiency.
- **Diode Switching Frequency**: The diodes in the front-end rectifier switch at the line frequency (e.g., 50/60 Hz). While this avoids high-frequency switching losses for these diodes, the overall conduction losses remain a concern.
- **Control Objectives**: The primary control objectives are to maintain a constant output voltage (Vo, typically greater than the peak input voltage Vs,pk) and ensure the input current is sinusoidal and in phase with the input voltage, thus achieving unity power factor. The duty ratio `d(t)` is varied across the line cycle to achieve these goals, following a pattern similar to `1 - |Vs,pk sin(ωt)| / Vo`.

2. The Standard Bridgeless PFC Converter

To address the limitations of the conventional boost PFC, the bridgeless PFC converter was developed. This topology eliminates the front-end diode bridge rectifier, integrating its functionality into the boost stage.

- **Architecture**: It typically uses four semiconductor devices: two diodes (D1, D2) and two active switches (S1, S2), often MOSFETs with their intrinsic body diodes. An inductor (L) and capacitor (C) complete the circuit.
- **Operation in Positive Half-Cycle (Vs > 0)**:
- During the `d(ts)` period (S1 is ON): Current flows from the source, through L, S1 (channel), the body diode of S2, and back to the source. The voltage across the inductor `VL` is approximately `Vs`.

- During the `(1-d)Ts` period (S1 is OFF): The inductor current forces diode D1 to conduct. Current flows from the source, through L, D1, the load (Vo), and returns via the body diode of S2 to the source. The voltage across the inductor `VL` is `Vs - Vo`.
- Crucially, switch S2 is not actively switched in the positive half-cycle; its body diode provides the return path. Diode D2 is also not in operation.
- **Operation in Negative Half-Cycle (Vs < 0)**:
- During the `d(ts)` period (S2 is ON): Current flows from the source, through the body diode of S1, S2 (channel), L, and back to the source. The voltage across the inductor `VL` is approximately `Vs`.
- During the `(1-d)Ts` period (S2 is OFF): The inductor current forces diode D2 to conduct. Current flows from the source, through the body diode of S1, the load (Vo), D2, L, and back to the source. The voltage across the inductor `VL` is `Vs - Vo`.
- Similarly, switch S1 is not actively switched in the negative half-cycle, and diode D1 is not in operation.
- **Advantages**:
- **Reduced Component Count**: Compared to the conventional boost PFC, it uses fewer discrete components for rectification.
- **Lower Conduction Losses**: At any given instant, only two semiconductor devices are in the conduction path (e.g., S1 and S2's body diode, or D1 and S2's body diode). This significantly reduces conduction losses compared to the three devices in the conventional boost PFC.
- **Disadvantage**:
- **High-Frequency Diode Switching**: Diodes D1 and D2 are required to switch at the high switching frequency (`fsw`). This necessitates the use of expensive fast recovery diodes and introduces significant reverse recovery losses, which can offset some of the efficiency gains from reduced conduction losses.

3. The Totem Pole PFC Converter

The Totem Pole PFC converter is an advanced bridgeless topology designed to mitigate the high-frequency diode switching issue of the standard bridgeless PFC. It achieves this by intelligently arranging the switches and diodes.

- **Architecture**: It consists of two legs. One leg is an "active half-bridge" (S1 and S2, typically MOSFETs with body diodes), and the other is a "diode-based half-bridge" (D1 and D2). The input AC source is connected between the inductor and the midpoint of the active half-bridge.

- **Operation in Positive Half-Cycle (Vs > 0)**:
- The D2 diode is always forward-biased and conducting throughout the entire positive half-cycle.
- During the `d(ts)` period (S2 is ON): Current flows from the source, through L, S2 (channel), and D2, returning to the source. The voltage across the inductor `VL` is `Vs`.
- During the `(1-d)Ts` period (S2 is OFF): The inductor current forces the body diode of S1 to conduct. To minimize losses, S1 is actively turned ON during this period. Current flows from the source, through L, S1 (channel), the load (Vo), and D2, returning to the source. The voltage across the inductor `VL` is `Vs - Vo`.
- In this half-cycle, S2 is switched with duty cycle `d(t)`, and S1 is switched with `1-d(t)`. D2 is continuously on, while D1 is off.
- **Operation in Negative Half-Cycle (Vs < 0)**:
- The D1 diode is always forward-biased and conducting throughout the entire negative half-cycle.
- During the `d(ts)` period (S1 is ON): Current flows from the source, through D1, S1 (channel), L, and back to the source. The voltage across the inductor `VL` is `Vs`.
- During the `(1-d)Ts` period (S1 is OFF): The inductor current forces the body diode of S2 to conduct. To minimize losses, S2 is actively turned ON during this period. Current flows from the source, through D1, the load (Vo), S2 (channel), L, and back to the source. The voltage across the inductor `VL` is `Vs - Vo`.
- In this half-cycle, S1 is switched with duty cycle `d(t)`, and S2 is switched with `1-d(t)`. D1 is continuously on, while D2 is off.
- **Key Advantages**:
- **Reduced High-Frequency Switching Losses**: The most significant advantage is that diodes D1 and D2 only switch at the line frequency (`fs`). This eliminates the need for expensive fast recovery diodes for D1 and D2 and drastically reduces reverse recovery losses, as `fsw` is much greater than `fs`.
- **High Efficiency**: By having only two devices in conduction at any time and minimizing high-frequency diode losses, the Totem Pole PFC achieves very high efficiency.
- **Component Selection**: Since D1 and D2 switch at line frequency, standard, slower recovery diodes can be used, reducing cost and complexity.

- **Duty Ratio**: The duty ratio `d(t)` variation for the active switches (S1 and S2) is the same as in the boost PFC and standard bridgeless PFC: `1 - |Vs,pk sin(ωt)| / Vo`. The specific switch (S1 or S2) that receives this duty ratio signal depends on the half-cycle.

Key Takeaways

- [ ] Conventional boost PFC converters suffer from high component count and conduction losses due to three devices conducting simultaneously.
- [ ] Bridgeless PFC converters reduce conduction losses by ensuring only two devices are in the current path at any given time.
- [ ] The standard bridgeless PFC topology requires diodes to switch at high frequency, leading to significant reverse recovery losses.
- [ ] The Totem Pole PFC converter is an advanced bridgeless topology that addresses the high-frequency diode switching problem.
- [ ] In Totem Pole PFC, active switches (S1, S2) operate at high switching frequency, while the diodes (D1, D2) operate at line frequency.
- [ ] This allows for the use of slower, less expensive diodes for D1 and D2, significantly improving overall efficiency by reducing reverse recovery losses.
- [ ] The duty ratio variation for bridgeless PFC converters is similar to that of the conventional boost PFC, ensuring unity power factor and output voltage regulation.
- [ ] Both bridgeless PFC variants offer advantages over the conventional boost PFC in terms of efficiency and component count.

Conclusion

The evolution from conventional boost PFC to bridgeless and then to Totem Pole PFC topologies represents a continuous drive towards higher efficiency and reduced power losses in power conversion systems. By strategically reconfiguring the power stage, bridgeless converters effectively minimize the number of conducting semiconductor devices, thereby cutting down conduction losses. The Totem Pole PFC, in particular, stands out by intelligently segregating high-frequency switching to active devices (MOSFETs) and relegating line-frequency operation to diodes, thus eliminating the need for costly and lossy fast-recovery diodes in the main power path. These advancements are critical for developing more efficient charging infrastructure, electric vehicle chargers, and other power electronics applications, contributing to energy conservation and system reliability.